Automotive IP & AI Security — Risk Tracking & Automated Escalation in Jira

A self-directed portfolio project demonstrating how a security/IP risk register with likelihood/impact scoring, standards mapping, and automated escalation can be built natively in Jira, extending the same automation-driven controls pattern used in the companion AI Project Controlling (EVM) project to a security and compliance domain.

Note: *This is a practice/demo build using fictional data, created to showcase Jira administration, automation, and dashboarding skills across a different domain (security/IP risk vs. cost/schedule performance). It is not tied to a real client or employer engagement.*

1. Problem Statement

Automotive IP and AI-system security programs need to track risk across multiple domains IP protection, AI model security, financial exposure, secure development practices, and regulatory compliance with consistent scoring and automatic escalation when risk crosses a threshold. This project explores building that register directly in Jira, using calculated risk scores and rule-based escalation rather than a separate GRC tool.

2. Project Structure

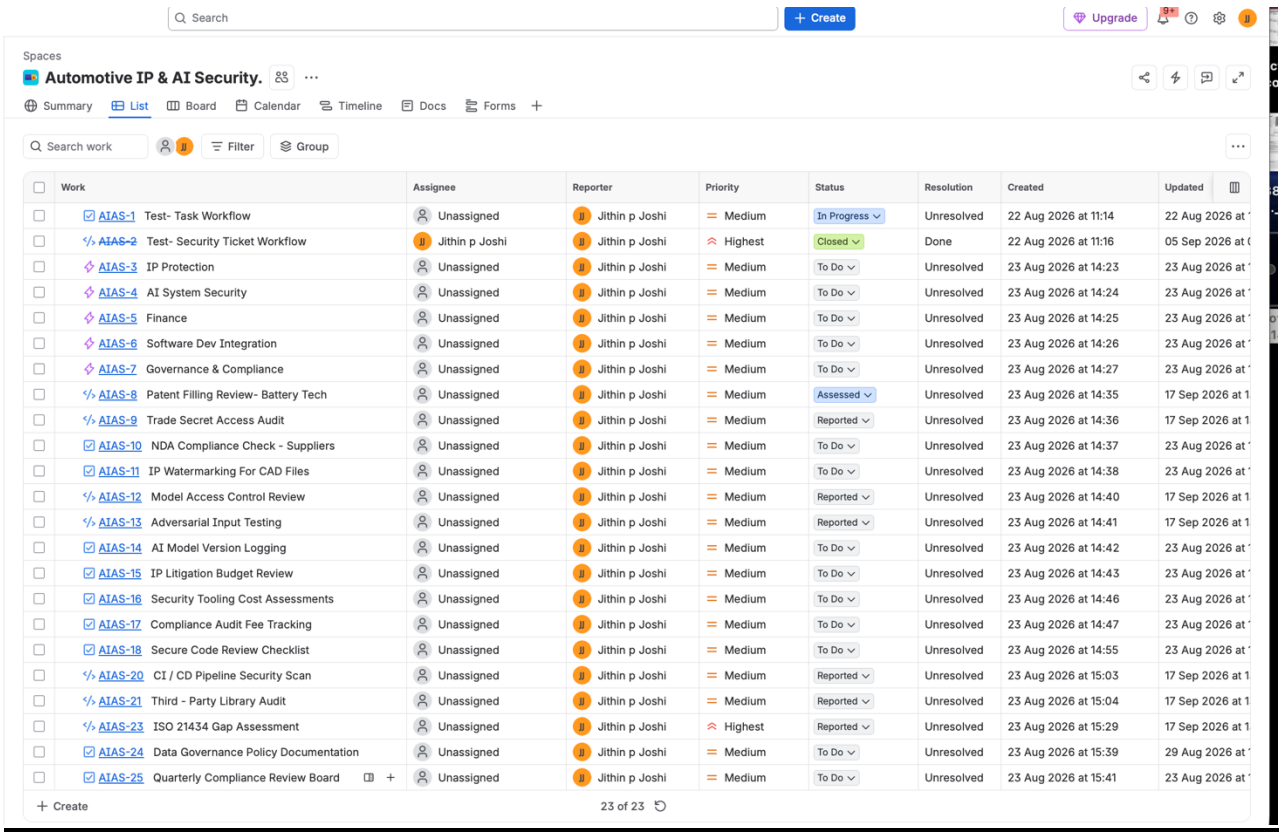
Space: Automotive IP & AI Security (Key: **AIAS**) — Team-managed Kanban.

3. Epics (risk domains):

Epic	Representative Tickets
IP Protection	Patent Filing Review, Trade Secret Access Audit, IP Litigation Budget Review, NDA Compliance Check, IP Watermarking.
AI System Security	Model Access Control Review, Adversarial Input Testing, AI Model Version Logging

Finance Breach Cost & Audit Trail	Compliance Audit Fee Tracking, Security Tooling Cost Assessments
Software Development Integration	CI/CD Pipeline Security Scan, Secure Code Review Checklist
Governance & Compliance	ISO 21434 Gap Assessment, Third-Party Library Audit, Data Governance Policy Documentation, Quarterly Compliance Review

*23 total issues across the 5 Epics, mixing Tasks and a custom **Security Ticket** work type.*



Full issue list: 5 Epics and 23 issues across Reported, To Do, In Progress, and Closed states.

4. Work Types & Workflow

- **Task** — standard workflow: **To Do → In Progress → Done**
- **Security Ticket** (custom work type) — custom workflow:
  Reported → Assessed → Mitigation In Progress → Resolved → Close

5. Custom Fields

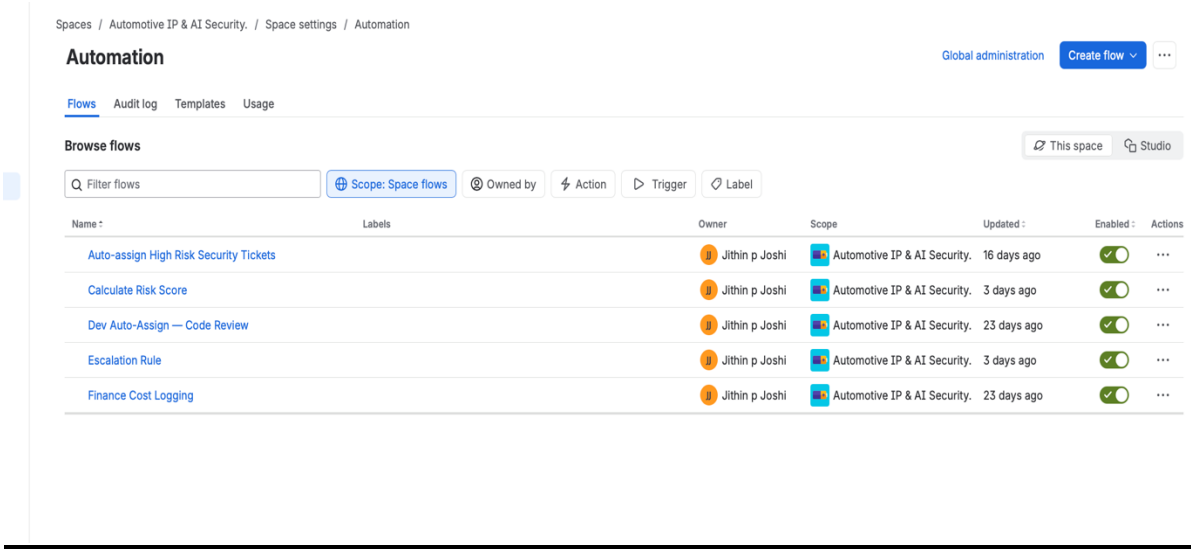
Field	Type	Purpose
Risk Category	Label	Classifies the nature of the risk
Affected System	Label	System or asset impacted
Standard Reference	Label (TISAX / ISO 21434 / GDPR)	Regulatory/standard mapping
Likelihood	Number (1–5)	Probability of occurrence
Impact	Number (1-5)	Severity if realized
Risk Score	Number (calculated)	Likelihood × Impact, set via automation
Estimated Cost	Number	Projected financial exposure
Requires Code Review (on Task)	Checkbox	Flags dev tasks needing security review

*Severity reuses Jira's system **Priority** field rather than a separate custom field.*

6. Automation Rules

Rule	Trigger	Condition	Action
Calculate Risk Score	Likelihood or Impact changed	_	Risk Score = Likelihood × Impact
Escalation Rule	Risk Score changed	Risk Score ≥ 15	Priority → Highest + escalation comment
Finance Cost Logging	Status → Resolved	Affected System = Financial Analytics	Auto-comment with Estimated Cost
Dev Auto-Assign — Code Review	Task created	Requires Code Review = true	Assign + post review checklist comment
Auto-assign High Risk Security Tickets	Risk Score changed	Risk Score ≥ threshold	Auto-assign to owner

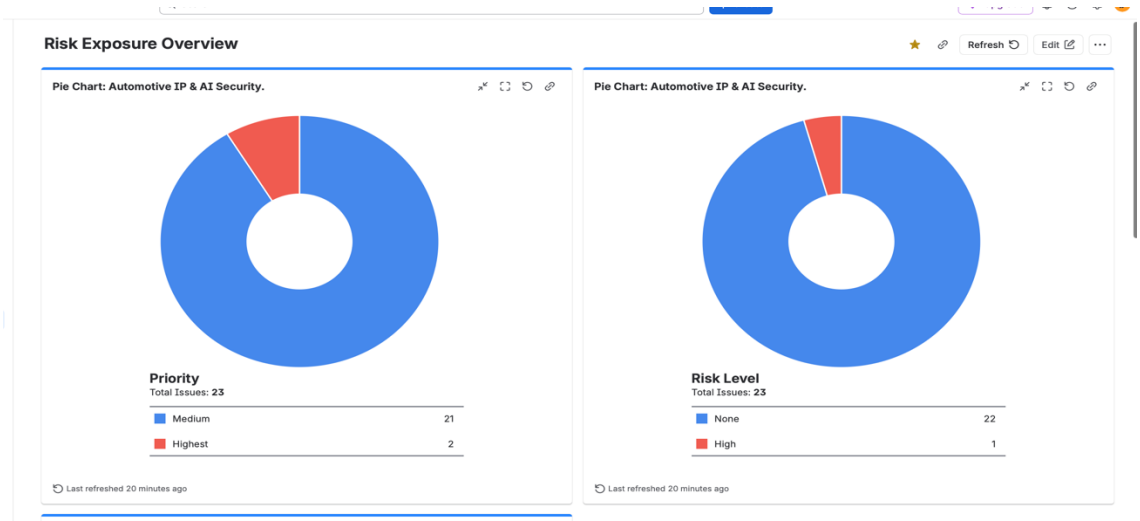
5 automation rules, all enabled



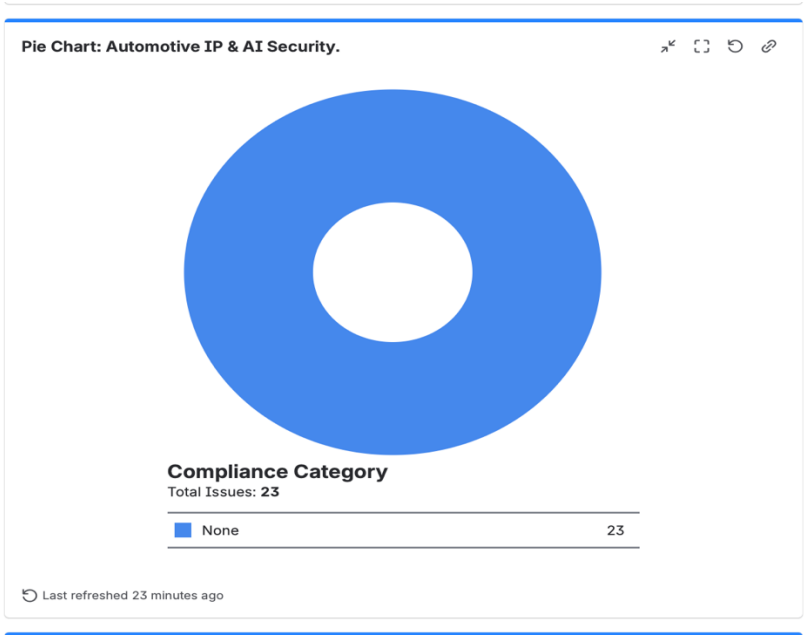
All 5 flows live in the space, owned and enabled.

7. Dashboard — "Risk Exposure Overview

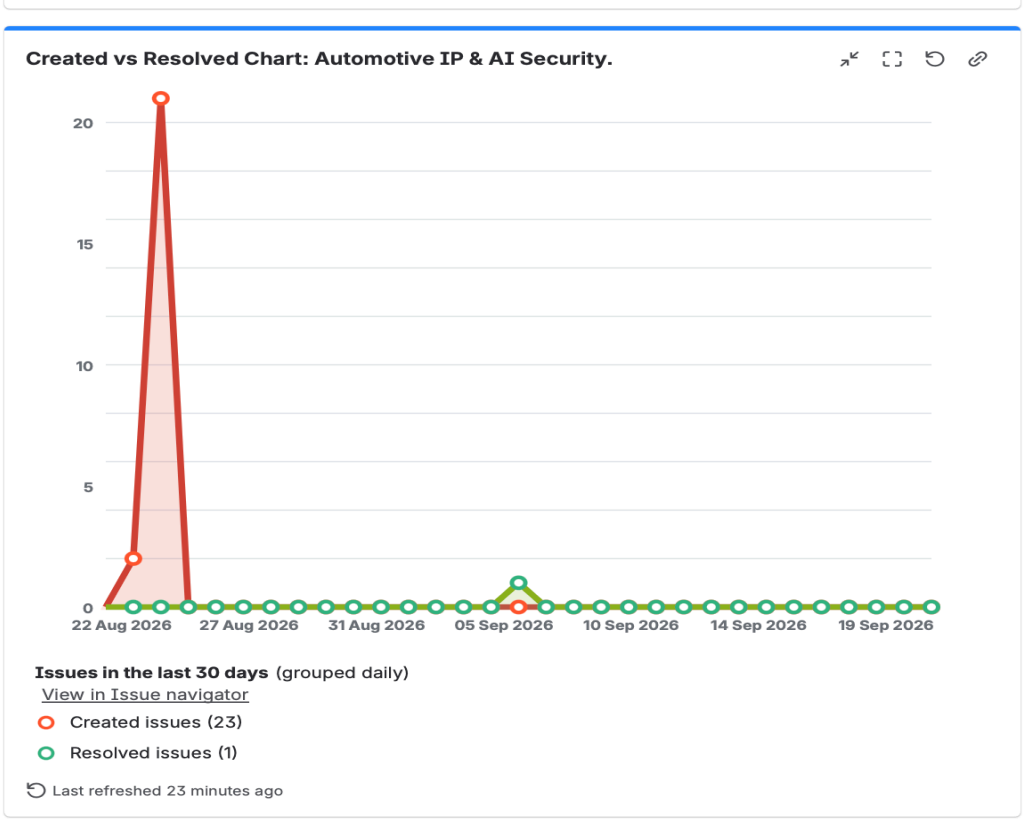
- Priority breakdown (pie chart)
- Risk Level breakdown (pie chart)
- Compliance Category breakdown (pie chart)
- Created vs Resolved chart



Priority breakdown (pie chart). Risk Level breakdown (pie chart)



Compliance Category breakdown (pie chart)



Created vs Resolved chart



Two Dimensional Filter Statistics: AIAS All Issues

Compliance Category	High	None	T:
None	1	22	23
Total Unique Issues:	1	22	23

Grouped by: Risk Level

Showing 1 of 1 statistics.

Last refreshed 22 minutes ago

Risk Level and Compliance Category distribution once all tickets carry live Likelihood, Impact, and Standard Reference values.

8. Known Limitation (Carried Forward from

As with the companion EVM project, Jira's "field value changed" trigger only fires on **future** edits — a value entered at ticket creation (or before the automation rule existed) does not retroactively trigger the rule. On first build, most Security Tickets showed no Risk Score and no Compliance Category value, because

- **Risk Score**: Likelihood/Impact were set once at creation, so the Calculate Risk Score rule never fired for most tickets.
- **Compliance Category (Standard Reference)**: this is a manual label field, not automation-driven — it was simply left unset on most tickets during initial buildout.

Resolution: each ticket's Likelihood/Impact was re-saved to force the Risk Score trigger, and Standard Reference was populated manually across all Security Tickets. This recurring pattern — catching an automation rule that silently does nothing because of trigger timing or a blank action field — is documented deliberately as a real, repeatable Jira automation pitfall.

9. Tech / Tools Used

- Jira (Team-managed Kanban space, custom work types, custom fields, automation engine, dashboards)

10. WhatThis Demonstrates

- Risk-scoring fundamentals (Likelihood × Impact matrices)
- Custom work type and workflow design in Jira
- Rule-based automation for scoring, escalation, and auto-assignment
- Cross-portfolio linking between related Jira spaces
- Debugging and correcting a recurring automation trigger-timing issue

11.Author

Built as a self-directed learning project — feedback welcome

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